week 11 30.03.2026

Approval-based committee elections

Setting:

- $N = \{1, \dots, n\}$ - set of voters
- $C = \{c_1, \dots, c_m\}$ - set of candidates
- $A_i \subseteq C$ - the approved set of voter $i \in N$
- The outcome is a committee $W \subseteq C$ such that $|W| = k$
- The utility of voter $i \in N$ for $W \subseteq C$ is $u_i(W) = |A_i \cap W|$.

Thiele's methods:

Given a sequence of weights $w_1, w_2, \dots$, select a committee W that maximises

$$\sum_{i \in N} \left(w_1 + w_2 + \dots + w_{u_i(W)} \right)$$

Examples:

- Approval voting (AV): $1, 1, 1, \ldots$
- Chamberlin-Courant (CC): $1, 0, 0, \ldots$
- Proportional approval voting (PAV): $1, \frac{1}{2}, \frac{1}{3}, \ldots$

That is,

- $\text{score}^{AV}(W) = \sum_{i \in N} |A_i \cap W|$

- $\text{score}^{CC}(W) = \sum_{i \in N} \mathbb{1}_{A_i \cap W \neq 0}$

  where $\mathbb{1}_{A_i \cap W \neq 0} = \begin{cases} 1 & A_i \cap W \neq 0 \\ 0 & A_i \cap W = 0 \end{cases}$

- $\text{score}^{PAV}(W) =$

$$\sum_{i \in N} \left(1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{|A_i \cap W|}\right)$$

Example: party lists

4 parties P_1, P_2, P_3, P_4 of 12 candidates each:

$$P_i \cap P_j = \emptyset \quad i \neq j$$

$$m_i = |P_i| = 12 \quad i = 1..4$$

$$\Rightarrow \boxed{m = 48}$$

P_1 is supported by 6 voters
P_2 ——— " ——— 4 voters
P_3 ——— " ——— 10 voters
P_4 ——— " ——— 2 voters

$$\Downarrow$$

$$\boxed{n = 22}$$

We want to select a committee of size

$$\boxed{k = 11}$$

with harmonic weights, we have:

P_1	P_2	P_3	P_4
+6	+4	+10	+2
+3	+2	+5	+1
+2	+1.33	+3.33	+0.66
+1.5	+1	+2.5	+0.5
+1.2	+0.8	+2	+0.4
+1	+0.66	+1.66	+0.33

Proportionality: Suppose a party list has x supporters with $x \geq \ell \cdot \frac{n}{K}$, then it deserves ℓ seats.

It holds that

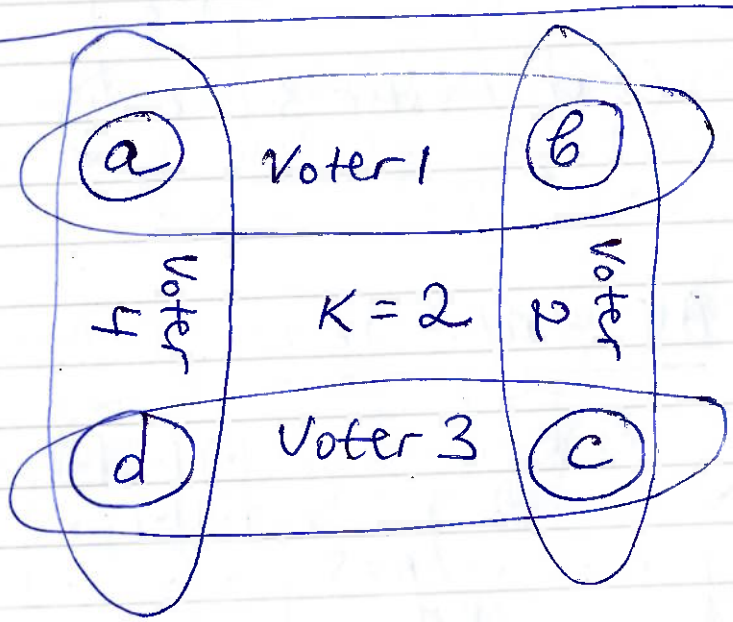
$$\frac{x}{1} > \frac{x}{2} > \dots > \frac{x}{\ell} \geq \frac{n}{K}$$

There can't be more than K alternatives with marginal increase at least $\frac{n}{K}$.

But how do we define proportionality when approval sets intersect?

First attempt:

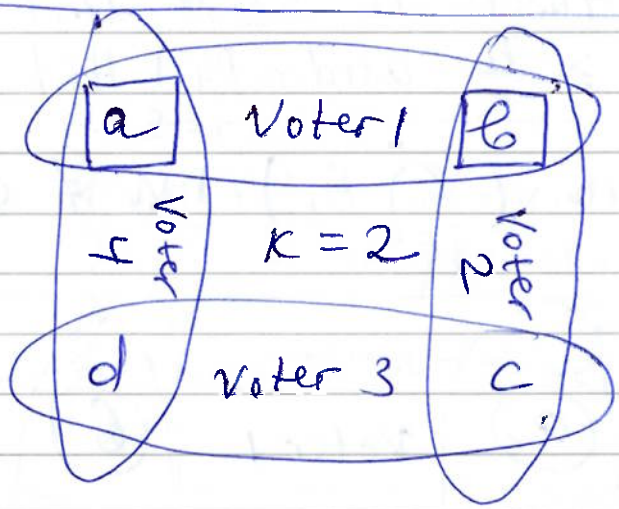
$$\boxed{\text{If there is } S \subseteq N \text{ such that } |S| \geq \frac{n}{K} \text{ and } \left|\bigcap_{i \in S} A_i\right| \geq 1 \text{ then } \left(\bigcap_{i \in S} A_i\right) \cap W \neq \phi.}$$



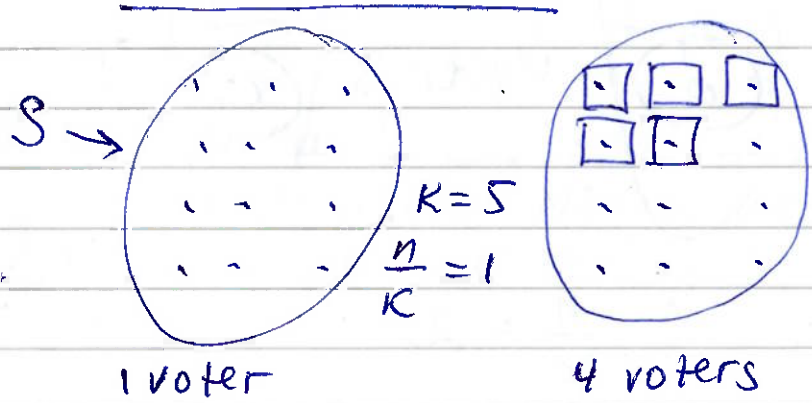
$$\frac{n}{K} = 2$$

Justified representation (JR)

If there is $S \subseteq N$ such that $|S| \geq \frac{n}{k}$ and $\left|\bigcap_{i \in S} A_i\right| \geq 1$ then $\exists i \in S$ such that $u_i(w) \geq 1$



AV fails JR:



Theorem: CC satisfies JR.

Proof: Let W be the CC committee violating JR, and let $S \subseteq N$ be the subset of voters witnessing the violation.

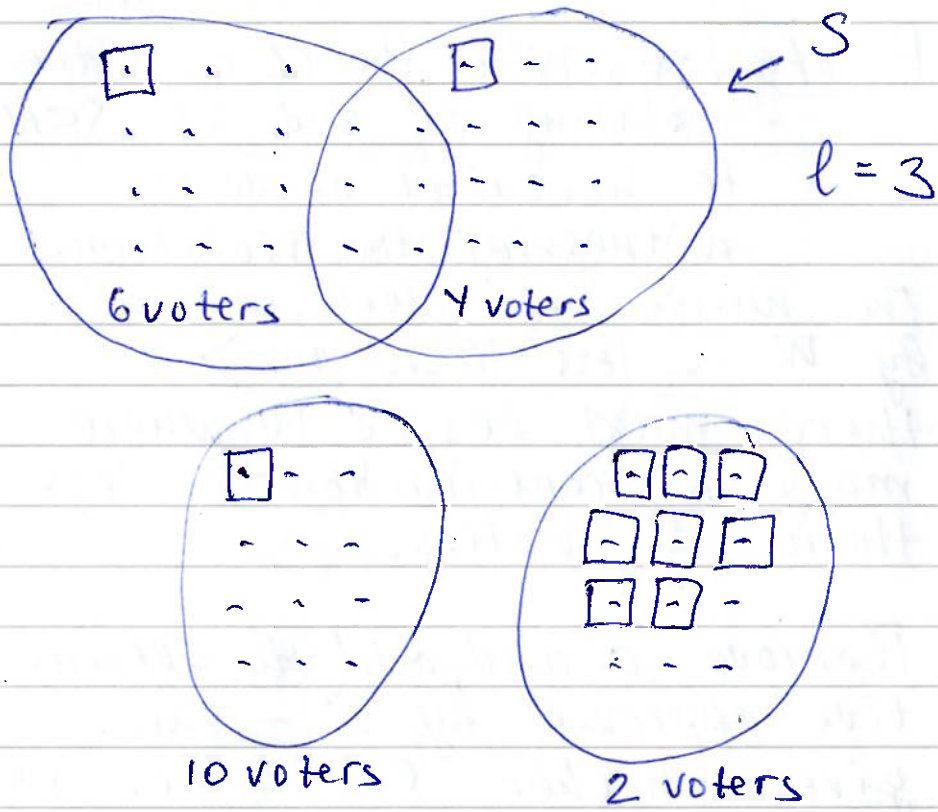
The number of voters covered by W is less than $n \Rightarrow$ there must be $c \in W$ whose marginal contribution is less than $\frac{n}{K}$ voters.

Remove c and add the alternative approved by S – this gives higher CC score. $\square$

Extended justified representation (EJR)

$$\boxed{\begin{array}{l}\text{If there is } S \subseteq N \text{ such that} \\ |S| \geq \ell \cdot \frac{n}{K} \text{ and } \left|\bigcap_{i \in S} A_i\right| \geq \ell \\ \text{then } \exists i \in S \text{ such that } u_i(W) \geq \ell\end{array}}$$

CC fails EJR:



$\ell = 3$

$n = 22 \quad k = 11 \quad \frac{n}{k} = 2$

EJR is clearly stronger than JR, so AV also fails EJR.

<u>Theorem:</u> PAV satisfies EJR.

<u>Proof:</u> Let W be the PAV committee, and suppose for contradiction that $S \subseteq N$ is such that $|S| \geq \ell \cdot \frac{n}{k}$ and $|\bigcap_{i \in S} A_i| \geq \ell$

but $u_i(W) < \ell$ for all $i \in S$.

Let $c^* \in \bigcap_{i \in S} A_i \setminus W$ and

$$W' = W \cup \{c^*\}.$$

Then,

$$\text{score}^{PAV}(W') \geq \text{score}^{PAV}(W) + |S| \cdot \frac{1}{\ell}$$

$$\geq \text{score}^{PAV}(W) + \frac{n}{k}$$

We claim that we can remove an alternative from W' and decrease the score^{PAV} by less than $\frac{n}{k}$

The average loss of score^{PAV} after removal is

$$\frac{1}{K+1} \sum_{c \in W'} \sum_{i : c \in A_i} \frac{1}{u_i(W')} =$$

$$\frac{1}{K+1} \sum_{i \in N} \sum_{c \in A_i \cap W'} \frac{1}{u_i(W')} \leq$$

$$\frac{1}{K+1} \sum_{i \in N} 1 < \frac{n}{K}$$

Hence, there is some $c' \in W$ such that

$$\text{score}^{PAV}(W' \setminus \{c'\}) > \text{score}^{PAV}(W)$$

in contradiction to the optimality of W. $\square$

Participatory budgeting

We now extend our discussion to a practical setting of participatory budgeting (PB) that is used by many cities to let their residents vote for projects that will be funded by the city government.

PB is a natural generalisation of approval-based committee elections, where instead of choosing a fixed-size committee, we select a subset of projects subject to a budget constraint. Voters approve the projects they support, and the goal is to choose an affordable set that reflects the preferences of the electorate as fairly as possible. In this setting, we shall also develop a rule that satisfies a proportionality

criterion in the spirit of extended justified representation ensuring that sufficiently large groups of voters receive appropriate influence over the final outcome.

<u>Model:</u>

- A set C of projects
- Each project $c \in C$ has a <u>cost</u> $cost(c) \geq 0$
- A set N of n voters
- Each voter $i \in N$ approves a subset $A_i \subseteq C$; we get a profile $P = (A_i)_{i \in N}$
- A total budget B
- Outcome: an outcome $W \subseteq C$ with

$$\sum_{c \in W} cost(c) \leq B$$

Notes:

- If $cost(c) = 1$ for all $c \in C$ and B is an integer, this is a committee election.

- For a set $T \subseteq C$ of projects, we write

$$cost(T) = \sum_{c \in T} cost(c).$$

Greedy method:

Almost all cities that use PB use the same voting method for deciding which projects win, which we call the Greedy Method.

1. $R \leftarrow C$, the remaining projects
2. $W \leftarrow \phi$
3. while $R \neq \phi$ do

- $c^* \leftarrow$ a project in R with the most votes, i.e. $c^* \in \arg\max_{c \in R} |\{i \in N : c \in A_i\}|$
- if $\sum_{c \in W} cost(c) + cost(c^*) \leq B$ then
  - $W \leftarrow W \cup \{c^*\}$
- $R \leftarrow R \setminus \{c^*\}$

4. Return W

What notion of welfare does the Greedy Method approximately maximize?

Recall the Knapsack problem: a list of items with payoffs $p_1, \ldots, p_m$ and costs $c_1, \ldots, c_m$, with an available budget B (the capacity of the Knapsack). The greedy algorithm for this problem selects projects in order of "bang-per-buck", i.e.

$\frac{p_i}{c_i}$, until the budget runs out.

Let $V_c = |\{i \in N : c \in A_i\}|$ be the vote count of project c.

The greedy method goes in order of

$$V_c = \frac{V_c \cdot cost(c)}{cost(c)}$$

Thus, the greedy method approximately solves a Knapsack problem with payoffs $V_c \cdot cost(c)$, and thus it approximately maximises

$$\sum_{i \in N} cost(A_i \cap W).$$

Based on this, one can argue that cities implicitly assume that voters have what are called <u>cost utilities</u>, i.e., their payoff is $u_i(W) = cost(A_i \cap W)$.

We could also use cardinality utilities with payoff

$$u_i(W) = |A_i \cap W|,$$

corresponding to selecting projects in order $v_c/cost(c)$.

In practice this leads to only the cheapest projects winning.

Advantages:

- Easy to understand for citizens and for government officials
- Approximately maximises the utilitarian social welfare $\left(\sum_{i \in N} u_i(W)\right)$

Disadvantages:

- Many voters may be

unrepresented ($W \cap A_i = \emptyset$)

- A large coordinated group of voters can get a disproportionately large share of the budget

- In practice, need to run separate elections for separate districts

- Small project have only a small chance of winning

- The method almost ignores the cost of projects

- There are incentives for project proposers to "bundle" their projects: if we merge two projects and thereby get more votes, we are more likely to win.

Proportional representation

Many of the disadvantages of the greedy method would be fixed by a rule that computes an outcome providing proportional representation, in the sense of satisfying an EJR-style property.

An outcome W satisfies Extended Justified Representation (EJR) if for all groups $S \subseteq N$ of voters and every proposal $T \subseteq C$ consisting of a set of projects that S can afford with its proportional share of the budget

$$\left(\text{cost}(T) \leq \frac{|S|}{|N|} \cdot B \right)$$

and that they all approve

$$\left(T \subseteq \bigcap_{i \in S} A_i\right),$$

there exists at least one voter $i \in S$ with

$$\text{cost}(A_i \cap W) \geq \text{cost}(T).$$

EJR will imply that cohesive groups of voters will not be unrepresented ($W \cap A_i = \phi$), and cannot be overridden by a larger cohesive group.

Consequence: Projects are guaranteed to win if they have enough single-minded supporters – if project c costs 20% of the budget and 20% of the voters only approve c, then c must win.

What rules satisfy EJR?

Natural first try: PAV! However, it is not obvious how to define it for cost utilities, and all natural definitions fail EJR.

There exist some other known ways to get EJR in the committee context, and many are polynomial time. But when projects have different costs, the following holds:

Theorem: Unless $P = NP$, there is no polynomial time algorithm that produces an EJR outcome.

This is because for the $n=1$ voter case, such an algorithm would need to solve SUBSET SUM.

There exists a method (computationally intractable) that always satisfies EJR.

But we can get "almost" EJR using natural polynomial time methods.

The Method of Equal Shares (MES)

1. Assign each voter $i \in N$ a virtual budget
$$b_i = \frac{B}{n}$$

2. $W \leftarrow \phi$

3. $R \leftarrow \{c \in C \setminus W : \sum_{i \in N : c \in A_i} b_i \geq cost(c)\}$,
the set of affordable projects.

4. If $R = \phi$ then return W

5. Select $c^* \in R$ with the highest "effective vote count"
(will define later, roughly: don't count voters with no money left)

6. $W \leftarrow W \cup \{c^*\}$

7. Deduct $cost(c^*)$ from the budgets of the voters of c^*, i.e. from $\{i \in N : c^* \in A_i\}$, spreading the cost as equally as possible (will define later)

8. Go to line 3.

Example: $\boxed{B = 1,100}$

project	cost	i_1	i_2	i_3	i_4	i_5	i_6	i_7	i_8	i_9	i_{10}	i_{11}	vote count
C_1	700	✓	✓	✓	✓	✓	✓					✓	7
C_2	400	✓	✓	✓	✓	✓	✓						6
C_3	250		✓		✓	✓		✓			✓		5
C_4	200							✓	✓	✓	✓		4
C_5	100							✓		✓	✓		3

Greedy selects C_1 and C_2.

MES:

$\forall i \in N, \quad b_i = \frac{1,100}{11} = 100$

$W = \phi$

$R = C$

Round 1:

$c^* = c_1$ with count 7

$W = \{c_1\}$

$b_1 = b_2 = b_3 = b_4 = b_5 = b_6 = b_{11} =$

$= 100 - \frac{700}{7} = 100 - 100 = 0$

c_2 and c_3 are no longer affordable

$R = \{c_4, c_5\}$

Round 2:

$C^* = C_4$ with count 4

$W = \{C_1, C_4\}$

$b_7 = b_8 = b_9 = b_{10} = 100 - \frac{200}{4} = 50$

$R = \{C_5\}$

Round 3:

$C^* = C_5$ with count 3

$W = \{C_1, C_4, C_5\}$

$b_7 = b_9 = b_{10} = 50 - \frac{100}{3} = 17$

$R = \phi \Rightarrow \boxed{W = \{C_1, C_4, C_5\}}$

Spreading cost as equally as possible

Sharing the cost of a project "as equally as possible" means that we distribute it among the project's supporters to

minimise the highest amount t that any of these voters contributes.

If we choose t optimally, every voter with more than t remaining (full voters) will contribute t, and all other voters (fractional voters) contribute whatever money they have left. If we sum up these contributions we cover precisely the cost of the project. Thus, if for every voter i we write b_i for their remaining money, we choose the (unique) value of t that satisfies

$$\underbrace{t + \dots + t}_{\text{full voters}} + \underbrace{b_{i_1} + \dots + b_{i_q}}_{\text{fractional voters}}$$

$$= \text{cost of the project}$$

$b_1 = \dots = b_5 = 100$

$b_6 = \dots = b_{10} = 20$

cost $(c) = 500$

Fractional voters contribute 20 each, and full voters contribute $t = \frac{500 - 20 \cdot 5}{5} = 80$ each.

<u>Effective vote count</u>

In order to compute the effective vote count of the project we first simulate how its cost would be shared: thus, we determine the value of t and the division into full and fractional voters as above.

The effective vote count of the project is then the number of full voters plus for each fractional vote i the fraction

of the full contribution t that i pays, that is b_i/t.

Note that voters who have no money left ($b_i = 0$) count as 0.

From the above equation, we see that

$$t \cdot \text{effective vote count} = \text{cost of the project}$$

In the above example with $b_1 \dots b_5 = 100$, $b_6 \dots b_{10} = 20$ and cost $(c) = 500$ we have voters 1..5 count as 1 each and voters 6..10 count as 0.25 each, so in total we have $5 \cdot 1 + 5 \cdot 0.25 = 6.25$

$t = 80$, so $80 \cdot 6.25 = 500$
$= \text{cost}(c)$

MES: more difficult example

In the earlier example with budget $B = 1,100$, projects $C = \{c_1, \dots, c_5\}$ with costs 700, 400, 250, 200, 100, respectively, and 11 voters $\{i_1, \dots, i_{11}\}$ assume that voter i_{11} stops approving c_1 and approves c_2 instead.

As before, we initially have $b_i = 100$ and $W = \emptyset$

However, project c_1 is no longer affordable, so $R = C \setminus \{c_1\}$.

Round 1:

$c^* = c_2$ with count 7

$W = \{c_2\}$

$b_1 = b_2 = b_3 = b_4 = b_5 = b_6 = b_{11} =$

$= 100 - \frac{400}{7} = 100 - 57 = 43$

$R = \{c_3, c_4, c_5\}$

What is the effective vote count of C_3 now?

We have 3 fractional supporters with $b_i = 43$, so the cost contributed by its 2 full supporters is $t = \frac{250 - 3 \cdot 43}{2} = 61$

and the corresponding effective vote count is

$$2 \cdot 1 + 3 \cdot \frac{43}{61} = 4.13$$

<u>Round 2:</u>

$c^* = C_3$ with count 4.13

$W = \{C_2, C_3\}$

$b_2 = b_4 = b_5 = 43 - 43 = 0$

$b_7 = b_{10} = 100 - 61 = 39$

$R = \{C_4, C_5\}$

What is the effective vote count of C_4 now?

There are 2 fractional voters with $b_i = 39$, so the cost contributed by the 2 full voters will be

$$t = \frac{200 - 2 \cdot 39}{2} = 61$$

and the corresponding effective vote count is

$$2 \cdot 1 + 2 \cdot \frac{39}{61} = 3.31$$

Round 3:

$c^* = C_4$ with count 3.31

$W = \{C_2, C_3, C_4\}$

$b_8 = b_9 = 100 - 61 = 39$

$b_7 = b_{10} = 39 - 39 = 0$

$R = \phi$

Final outcome

MES satisfies EJR up to one project

An outcome W satisfies EJR up to one project if for all groups $S \subseteq N$ of voters and every proposal $T \subseteq C$ consisting of a set of projects that S can afford

$$\left(\text{cost}(T) \leq \frac{|S|}{|N|} \cdot B \right)$$

and that they all approve

$$\left(T \subseteq \bigcap_{i \in S} A_i \right)$$

we have that either

$$T \subseteq W$$

or there exists $i \in S$ and a project $p^* \in T \setminus W$ with

$$\text{cost}((A_i \cap W) \cup \{p^*\}) \geq \text{cost}(T)$$